CURRICULUM VITAE
DR. SIMONETTA GRILLI

Current Position

Director of Research at Institute of Applied Sciences and Intelligent Systems (**CNR-ISASI**)

Personal Information

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Executive Summary

Dr. Simonetta Grilli is Director of Research at the Institute of Applied Sciences and Intelligent Systems "Eduardo Caianiello" of the National Research Council (CNR-ISASI, Pozzuoli), where for over 20 years she has been conducting research at an international level on the following main topics:

- **Applied Optics & Quantitative Microscopy**
  Interferometric and holographic modalities for label-free, 3D characterization and quantitative analysis of biological and physical samples; manipulation of matter at the micro- and nanoscale through interference lithography and laser structuring
- **Light-matter Interactions & Cellular Biophotonics**
  Investigation of light induced phenomena in biological samples; live-cell imaging and impact of coherent light on cellular viability; high-contrast and label-free visualization of intracellular features
- **Functional Materials, Soft Matter**
  Non-contact manipulation, micro-patterning, and actuation of soft matter, viscoelastic fluids, and polymeric films via localized thermal and electric gradients
- **Pyro-electrohydrodynamics & High-Sensitivity Biosensing Platforms**
  Pioneering the use of pyroelectrically induced electric fields for high-precision printing of biological samples; innovative platforms for sub-picogram detection of neurological biomarkers (e.g. Tau proteins and amyloid aggregates); plasmonic- and fluorescence-based nanostructures for biosensing applications

Bibliometrics

- **Lista of Publications:** The complete bibliometric output is accessible via https://scholar.google.com/citations?user=rM9jPGkAAAAJ&hl=it . An extended list of representative main publications is attached as **Annex 1**.
- **Total Publications:** >200 peer-reviewed articles in international journals, book chapters, and proceedings
- **Citations:** >4,900 citations
- **h-index: 44** (Google Scholar) / **41** (Scopus)
- **High-Impact Journal Portfolio:** *Nature Nanotechnology*, *Nature Communications*, *PNAS*, *Advanced Materials*, *Advanced Functional Materials*, *ACS Applied Materials & Interfaces*, *Optica*, *Physical Review Letters*.

Education & Academic Qualifications

Year	Degree/Qualification	Institution	Details/Specialization
2012	Italian National Scientific Qualification (ASN)	MIUR (Italy)	Qualified as Associate Professor in Experimental Condensed Matter Physics (02/B1)
2006	Ph.D. in Physics	KTH Royal Institute of Technology, Stockholm (Sweden)	Thesis: *"Ferroelectric domain engineering and characterization for photonic applications"* (ISBN 91-7178-382-2). Focused on holographic techniques, optical diagnostics, and real-time poling monitoring.
1999	Master's Degree in Physics	University of Naples "Federico II", Italy	100/100 cum laude. Thesis on optical characterization of biological tissue mimetics for optical mammography applications.

Academic Teaching, Supervision & Educational Leadership

- **Lecturer** for *"Pyro-electrohydrodynamics and Advanced Technologies for Soft Matter Manipulation"* (4 ECTS, 16 Hours) within the PhD Program in Mathematics, Physics and Applications for Engineering, University of Campania "L. Vanvitelli"
- Member of the **Teaching Board** in PhD Program in Mathematics, Physics and Applications for Engineering, University of Campania "L. Vanvitelli"
- Member of the **Self-Evaluation Committee** of the PhD Program in Mathematics, Physics and Applications for Engineering at the University of Campania "L. Vanvitelli"
- Member of the **Steering Committee** of the Mathematics Degree Programs at the University of Campania “L. Vanvitelli
- **Supervision of PhD students** in Physics and Materials Engineering across the University of Campania (Department of Mathematics and Physics) and University of Naples (DICMaPI).
- **Supervision of Master students** in Physics, Bioengineering and Chemical Engineering theses at University of Naples "Federico II"
- **International Internships** with supervision of visiting European PhDs and interns (e.g., Vrije Universiteit Brussel).
- **School Leadership:** Co-Director and Co-Organizer of the International Academic School *"Nano-Tooling"* (2017), training early-stage researchers in nano-photonics and microfluidics.

Main Research Grants & Project Leadership

Role	Project Title & Grant Code	Funding Agency / Program	Period
Project Coordinator	SensApp *(Super-sensitive detection of Alzheimer's disease biomarkers)* — GA 829104	European Commission (H2020 FET-Open)	2019 – 2022
Project Coordinator	Nano-Bio-Guard *(Portable pyro-electro-hydrodynamic biosensor)*	MIUR - FIRB Futuro in Ricerca	2012 – 2016
Senior Investigator	Space Health Technologies	Italian Space Agency (ASI)	2021 – 2025

Role	Project Title & Grant Code	Funding Agency / Program	Period
Partner / Work Package Leader	Real3D *(Real-Time 3D Color Holographic Displays)*	European Commission (FP7)	2008 – 2011
Scientific Chair	ABITAS *(Advanced Optical Diagnostics for Surfaces)*	European Commission (FP6)	2004 – 2007
Principal Investigator	Industrial R&D Contracts	Heinz S.p.A., ITP S.r.l., IRCCS Bonino Pulejo	2015 – 2022

Academic Honors, Awards & International Recognition

- **SPIE Fellow (2023):** Elected Fellow of the International Society for Optics and Photonics (SPIE) for achievements in biophotonics, pyro-electrohydrodynamics, and optical manipulation.
- **EC Innovation Radar Award (2020):** SensApp awarded by the European Commission for its innovation value
- **IEEE-LEOS Doctoral Thesis Award (2006):** Global recognition for outstanding doctoral research in electro-optics and ferroelectric domain engineering.
- **Plenary & Invited Speaker:** Plenary speaker at *SPIE Photonics Europe* (Strasbourg, 2006), and invited speaker at >30 international conferences (CLEO, SPIE Optics+Photonics, IEEE IPC).
- **Cover Features & Highlights:** publication selected for *Optics & Photonics News* "Best of Optics" highlights in 2008, 2011, and 2012; featured on the Front Cover of *Proceedings of the National Academy of Sciences (PNAS)* (2011); publication selected for *Wiley Hot Topics* in Materials Science (2012); research highlighted in *Nature Materials* and *Nature Nanotechnology* News & Views.

Professional Service, Evaluation & Institutional Governance

- **CNR Technical Scientific Committee (2025 – 2026):** Head of the internal evaluation panel for PRIN 2022 PNRR projects in PE7 area (Physical Sciences and Engineering).
- **CNR Selection Boards:** President of Board for Research Directors (call CNR315.62, area Biophysics); Board Member for Senior Researchers; President of Board for Postdoctoral Fellows.
- **Expert Evaluator for the European Commission:** Reviewer and evaluator for projects submitted to FP6-2005-IST-5 call.
- **Editorial Board Member:** *JPhys Photonics* (IOP Publishing) and *Optica* publishing groups.
- **Co-editor** of the Springer book “Ferroelectric Crystals for Photonic Applications” (Springer Verlag, 2008).
- **Author of the monograph** “Ferroelectric Domain Engineering and Characterization for Photonic Applications” (2006).
- **Conference Co-Chair:** Biennial SPIE International Conference *"Optical Methods for Inspection, Characterization, and Imaging of Biomaterials"* (Munich, Germany, 2019 – Present).
- **Advisory Board Member:** EU Project 360 CARLA (2024 – Present), focusing on photonics career development across Europe.
- **Scientific Committee:** Member of SC of MOC 2026 conference.

- **Institutional Delegate:** Delegate of the CNR President to the Campania Aerospace District (DAC S.c.a.r.l.) (2023 – Present).
- **Peer Reviewer:** Regular reviewer for premier scientific journals, including *Nature Photonics*, *ACS Nano*, *Advanced Materials*, *Optics Express*, *Applied Physics Letters*, and *IEEE Transactions on Biomedical Engineering*.

Career Breaks

- **Maternity Leave:** December 2008 – April 2009
- **Maternity Leave:** March 2011 – July 2011

ANNEX 1

Dr. Simonetta Grilli – Main Publications
(Listed in non-chronological order)

Whole optical wavefields reconstruction by digital holography
S Grilli, P Ferraro, SD Nicola, A Finizio, G Pierattini, R Meucci
Optics Express 9 (6), 294-302, 2001

Dispensing nano–pico droplets and liquid patterning by pyroelectrodynamic shooting
P Ferraro, S Coppola, S Grilli, M Paturzo, V Vespini
Nature nanotechnology 5 (6), 429-435, 2010

Liquid micro-lens array activated by selective electrowetting on lithium niobate substrates
S Grilli, L Miccio, V Vespini, A Finizio, S De Nicola, P Ferraro
Optics Express 16 (11), 8084-8093, 2008

Quantitative phase microscopy of microstructures with extended measurement range and correction of chromatic aberrations by multiwavelength digital holography
P Ferraro, L Miccio, S Grilli, M Paturzo, S De Nicola, A Finizio, R Osellame, ...
Optics express 15 (22), 14591-14600, 2007

Electro-drawn drug-loaded biodegradable polymer microneedles as a viable route to hypodermic injection
R Vecchione, S Coppola, E Esposito, C Casale, V Vespini, S Grilli, ...
Advanced Functional Materials 24 (23), 3515-3523, 2014

3D lithography by rapid curing of the liquid instabilities at nanoscale
S Grilli, S Coppola, V Vespini, F Merola, A Finizio, P Ferraro
Proceedings of the National Academy of Sciences 108 (37), 15106-15111, 2011

Reversible holographic patterns on azopolymers for guiding cell adhesion and orientation
C Rianna, A Calabuig, M Ventre, S Cavalli, V Pagliarulo, S Grilli, ...
ACS Applied Materials & Interfaces 7 (31), 16984-16991, 2015

Effects of lithium niobate polarization on cell adhesion and morphology
V Marchesano, O Gennari, L Mecozzi, S Grilli, P Ferraro
ACS Applied Materials & Interfaces 7 (32), 18113-18119, 2015

Surface-charge lithography for direct PDMS micro-patterning
S Grilli, V Vespini, P Ferraro
Langmuir 24 (23), 13262-13265, 2008

Wettability patterning of lithium niobate substrate by modulating pyroelectric effect to form microarray of sessile droplets
P Ferraro, S Grilli, L Miccio, V Vespini
Applied Physics Letters 92 (21), 2008

Active accumulation of very diluted biomolecules by nano-dispensing for easy detection below the femtomolar range
S Grilli, L Miccio, O Gennari, S Coppola, V Vespini, L Battista, P Orlando, ...
Nature Communications 5 (1), 5314, 2014

Surface nanoscale periodic structures in congruent lithium niobate by domain reversal patterning and differential etching
S Grilli, P Ferraro, P De Natale, B Tiribilli, M Vassalli
Applied Physics Letters 87 (23), 2005

Investigating fibroblast cells under “safe” and “injurious” blue-light exposure by holographic microscopy

A Calabuig, M Mugnano, L Miccio, S Grilli, P Ferraro
Journal of Biophotonics 10 (6-7), 919-927, 2017

Light induced DEP for immobilizing and orienting Escherichia coli bacteria
L Miccio, V Marchesano, M Mugnano, S Grilli, P Ferraro
Optics and Lasers in Engineering 76, 34-39, 2016

In situ investigation of periodic poling in congruent LiNbO3 by quantitative interference microscopy
S Grilli, M Paturzo, L Miccio, P Ferraro
Measurement Science and Technology 19 (7), 074008, 2008

Investigation on reversed domain structures in lithium niobate crystals patterned by interference lithography
S Grilli, P Ferraro, S De Nicola, A Finizio, G Pierattini, P De Natale, ...
Optics Express 11 (4), 392-405, 2003

In vitro cytotoxicity evaluation of cadmium by label-free holographic microscopy
M Mugnano, P Memmolo, L Miccio, S Grilli, F Merola, A Calabuig, ...
Journal of Biophotonics 11 (12), e201800099, 2018

Investigation on cone jetting regimes of liquid droplets subjected to pyroelectric fields induced by laser blasts
O Gennari, L Battista, B Silva, S Grilli, L Miccio, V Vespini, S Coppola, ...
Applied Physics Letters 106 (5), 2015

Spiral formation at the microscale by μ-pyro-electrospinning
L Mecozzi, O Gennari, R Rega, S Grilli, S Bhowmick, MA Gioffrè, ...
Soft Matter 12 (25), 5542-5550, 2016

A skin-over-liquid platform with compliant microbumps actuated by pyro-EHD pressure
O Gennari, R Rega, M Mugnano, E Oleandro, L Mecozzi, V Pagliarulo, E Mazzon, A Bramanti, A Vettoliere, C Granata, P Ferraro, S Grilli
NPG Asia Materials 11 (1), 1, 2019

Control of lateral domain spreading in congruent lithium niobate by selective proton exchange
S Grilli, C Canalias, F Laurell, P Ferraro, P De Natale
Applied Physics Letters 89 (3), 2006

In situ visualization of domain kinetics in flux grown KTiOPO4 by digital holography
C Canalias, V Pasiskevicius, F Laurell, S Grilli, P Ferraro, P De Natale
Journal of Applied Physics 102 (6), 2007

Label-free quantification of the effects of lithium niobate polarization on cell adhesion via holographic microscopy
B Mandracchia, O Gennari, A Bramanti, S Grilli, P Ferraro
Journal of Biophotonics 11 (8), e201700332, 2018

Pyroelectric effect enables simple and rapid evaluation of biofilm formation
O Gennari, V Marchesano, R Rega, L Mecozzi, F Nazzaro, F Fratianni, R Coppola, L Masucci, E Mazzon, A Bramanti, P Ferraro. S Grilli
ACS applied Materials & Interfaces 10 (18), 15467-15476, 2018

Maskless arrayed nanofiber mats by bipolar pyroelectrospinning
R Rega, O Gennari, L Mecozzi, V Pagliarulo, A Bramanti, P Ferraro, S Grilli
ACS applied Materials & Interfaces 11 (3), 3382-3387, 2019

Plasmon resonance of gold nanorods for all-optical drawing of liquid droplets
M De Angelis, P Matteini, F Ratto, R Pini, S Coppola, S Grilli, V Vespini, ...
Applied Physics Letters 103 (16), 2013

High-rate accumulation of tiny aqueous droplets using a pyroelectrohydrodynamic jet system
V Tkachenko, R Schwödiauer, R Rega, S Itri, M Mugnano, D del Giudice, H Ottevaere, Y Nie, P Ferraro, S Grilli

Advanced Engineering Materials 24 (3), 2100756, 2022

Direct printing of gold nanospheres from colloidal solutions by pyro-electrohydrodynamic jet allows hypersensitive SERS sensing
M Pannico, P Musto, R Rega, V Vespini, S Grilli, P Ferraro
Applied Surface Science 531, 147393, 2020

Highly sensitive detection of the neurodegenerative biomarker Tau by using the concentration effect of the pyro-electrohydrodynamic jetting
C Di Natale, S Coppola, V Vespini, V Tkachenko, S Russo, G Luciani, G Vitiello, F Ferranti, S Mari, P Ferraro, PL Maffettone, S Grilli
Biosensors and Bioelectronics 254, 116234, 2024

High-accuracy characterization of pyroelectric materials: A noncontact method based on surface potential measurements
R Schwödiauer, S Bauer-Gogonea, M Kaltenbrunner, V Tkachenko, S Grilli
Journal of Advanced Dielectrics 13 (04), 2341002, 2023

Characterization of bovine serum albumin immobilization on surface modified glass slides in case of pyro-electrohydrodynamic spots
C Di Natale, S Coppola, V Vespini, V Tkachenko, G Luciani, G Vitiello, F Ferranti, S Mari, PL Maffettone, S Grilli
Analytica Chimica Acta 1329, 343178, 2024

Enhancing the performance of planar SERS sensors with pyroelectrohydrodynamic jet printing
M Pannico, V Vespini, V Tkachenko, S Coppola, F Graziano, P Musto, P Ferraro, S Grilli
ACS Applied Materials & Interfaces 17 (39), 55491, 2025

An All-Optical Driven Bio-Photovoltaic Interface for Active Control of Live Cells
L Miccio, M Mugnano, V Marchesano, A Calabuig, Z Wang, G Giordano, F Borrelli, S Grilli, Ferraro
Advanced Functional Materials 36 (39), e18941, 2026

Highly sensitive detection of different aggregation stages of Aβ1-42-amyloid peptide in urine-like samples by auto-fluorescence
C Di Natale, S La Manna, S Carbone, S Coppola, V Vespini, V Tkachenko, D Tammaro, S Russo, G Vitiello, G Luciani, D Marasco, PL Maffettone, P Ferraro, S Grilli
Analytica Chimica Acta, 344531, 2025

Surface-confined FRET nanoplatform printed via pyro-EHD jet for stable and reproducible TNF-α detection
S Carbone, S Russo, A Palma, GJ Mosca, S La Manna, A Cugudda, S Coppola, PL Maffettone, S Grilli, G Vitiello, C Di Natale
RSC Advances 16 (22), 20304-20315, 2026

Fluorescence spectroscopy of scattering media in visible and infrared range
M Lepore, S Grilli, PL Indovina
Optical Biopsy and Tissue Optics 4161, 180-189, 2000